

Problem: John invests at 10% simple interest. He invests 1000 dollars. How much does he have after 3 years?

Learner persona: High School, grade 11-12 (age 16-18)

Step-DPO (baseline)

Step 1: First, calculate the annual interest earned by John. Since he invests 1000 at a 10% rate, the annual interest is $1000 \times 0.10 = 100$ dollars.

Step 2: Determine the total interest over 3 years. Multiply the annual interest by the number of years: $100 \times 3 = 300$ dollars.

Step 3: Add the total interest to the principal: $1000 + 300 = 1300$ dollars.

Final answer: 1300

Computes year-by-year without a formula - same as a younger learner; not adapted to the level.

SLC-StepDPO (Ours)

Step 1: Calculate the simple interest. The formula for simple interest is $I = P \times r \times t$, where P is the principal, r the annual rate, and t the time in years. With $P = 1000$, $r = 0.10$, $t = 3$: $I = 1000 \times 0.10 \times 3 = 300$.

Step 2: Add the interest to the principal. The total amount is $A = P + I = 1000 + 300 = 1300$.

Final answer: 1300

States the general formula $I = P \times r \times t$ with formal notation - concise and level-appropriate.

Figure. Same problem for an advanced high-school learner (greedy decoding, real outputs). Red = under-adapted (no formula); Green = formal, level-appropriate formula. Both reach 1300.